

Data Management

Hao Ji

Data Scientist

CARC

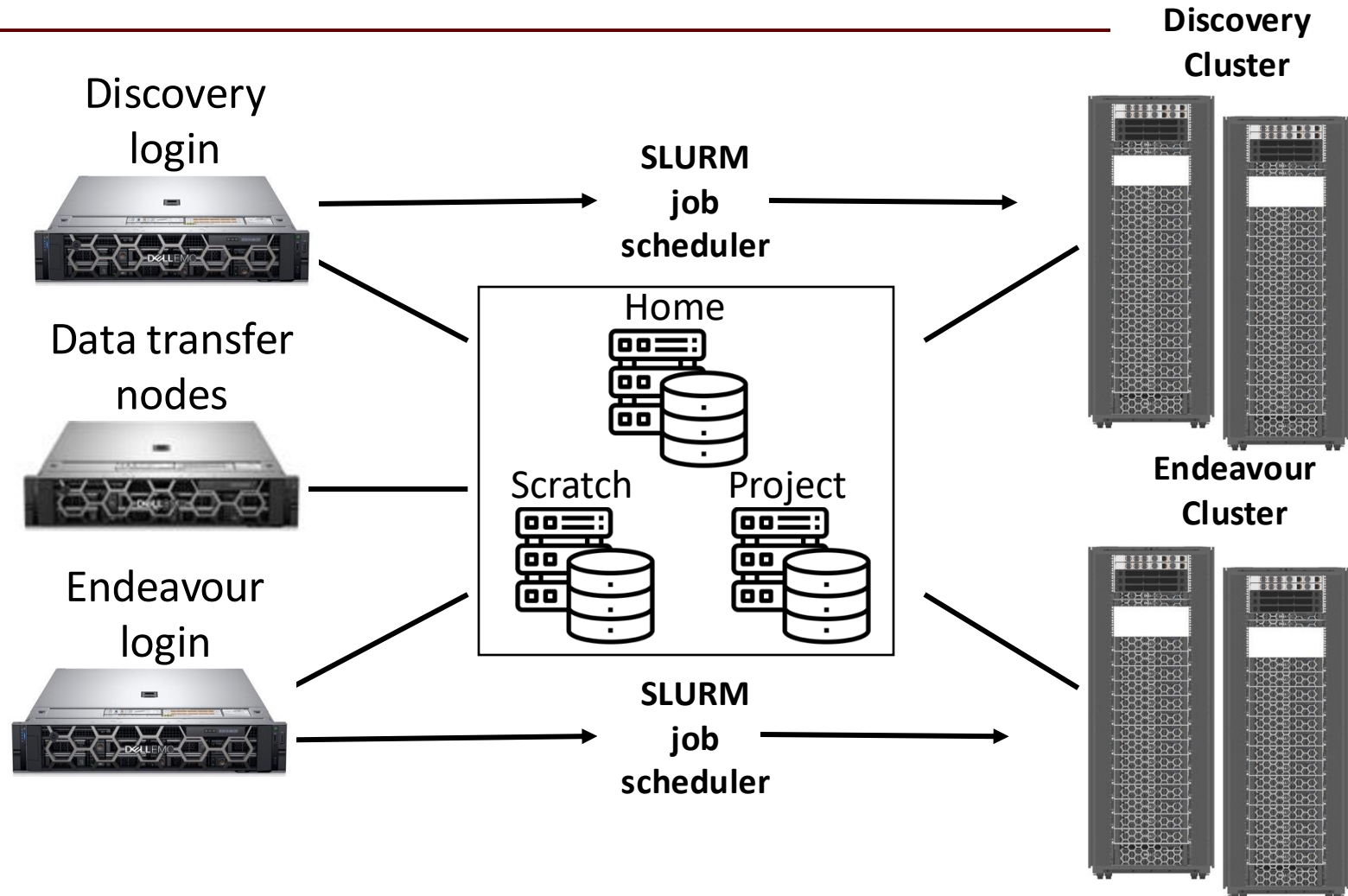


USC

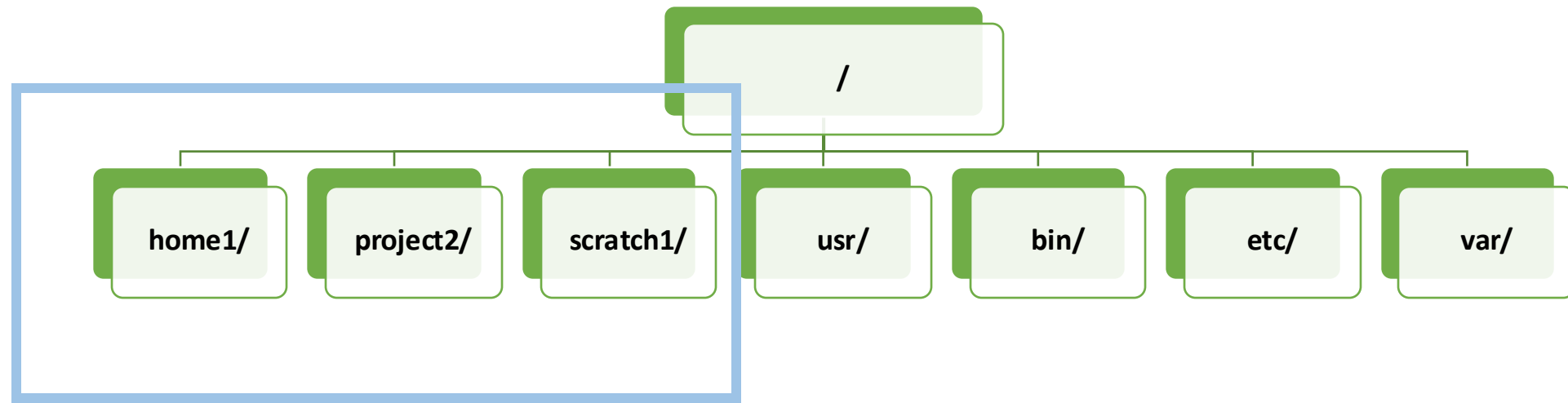
Advanced Research Computing
Enabling scientific breakthroughs at scale

CARC Computing Resources

- Head nodes (Login nodes)
 - e.g., discovery.usc.edu
- Compute nodes
 - 510 nodes running Rocky
- Networked storage
 - Disk arrays and file servers
- Low-latency network
- SLURM resource scheduler



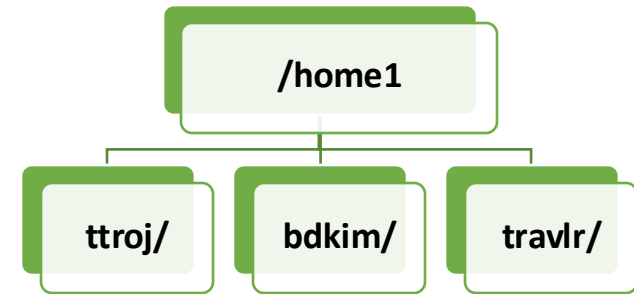
CARC Filesystem



Home (private)

- Users login to their home directory
 - Personal, private directory
 - Only user can modify files here
- User quotas
 - 100 GB of disk quota & 255K files

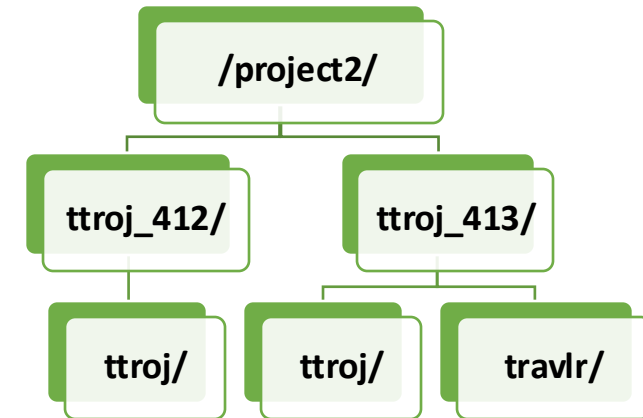
`/home1/<user_name>`



Project Directory

- Projects may have dedicated storage
 - Permissions and structure determined by each group
- Up to 10TB per PI free
 - Shared among group members
- Additional storage available
 - Increments of 5TB
 - \$60/ TB / year

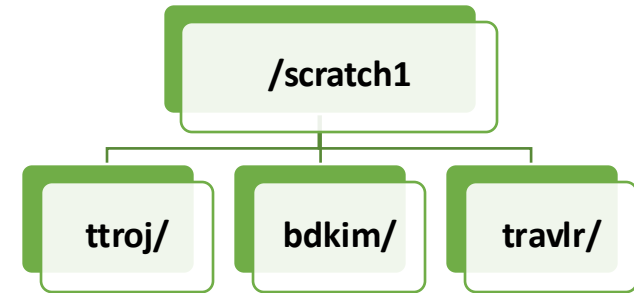
`/project2/<pi_username>_<id>`



Scratch directories

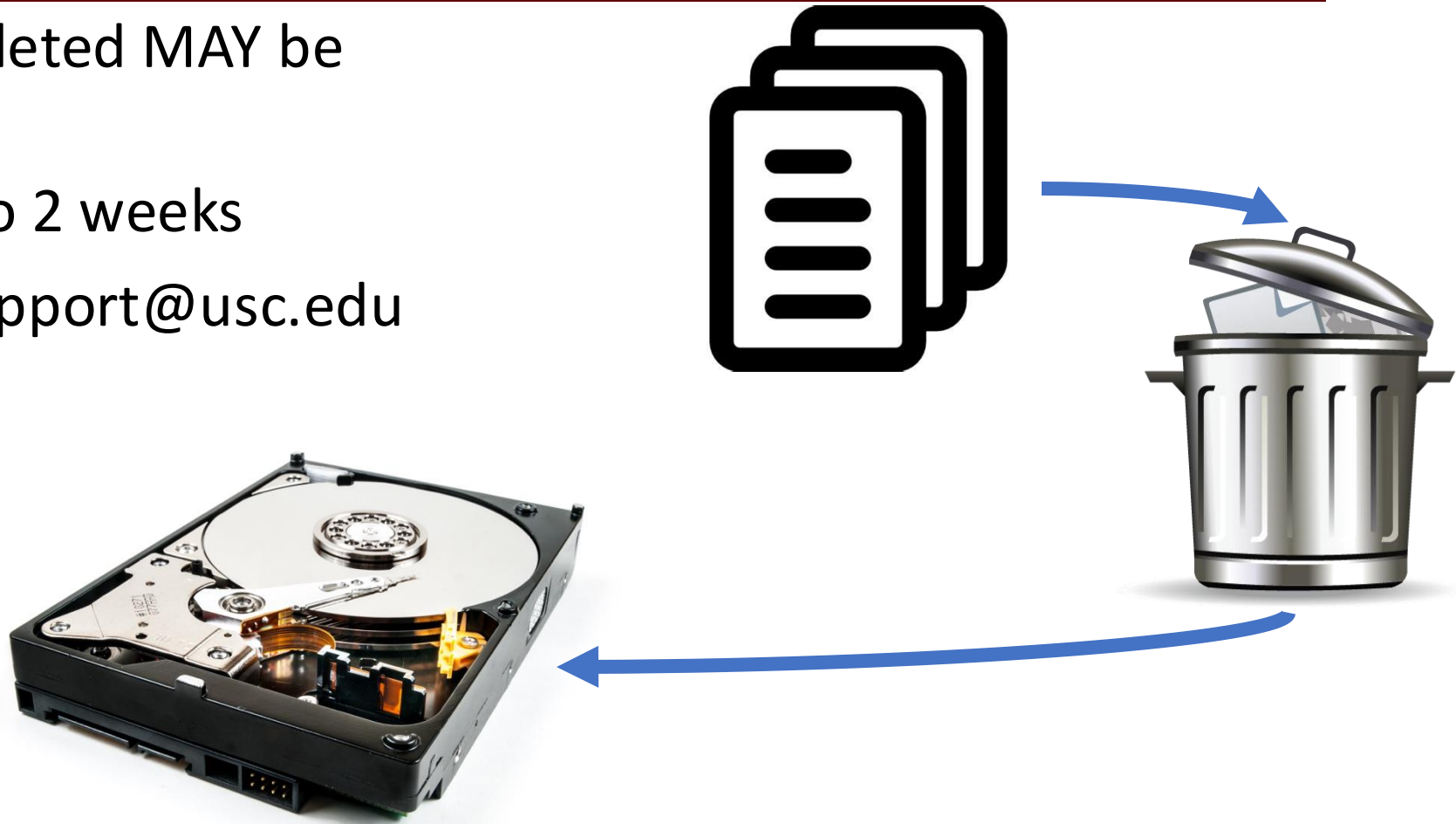
- By default, only accessible by owner
- Not backed up
 - copy important files elsewhere

`/scratch1/<user_name>`



Data Recovery

- Data accidentally deleted MAY be recovered
- Snapshots held up to 2 weeks
- Send mail to carc-support@usc.edu



Uploading Data



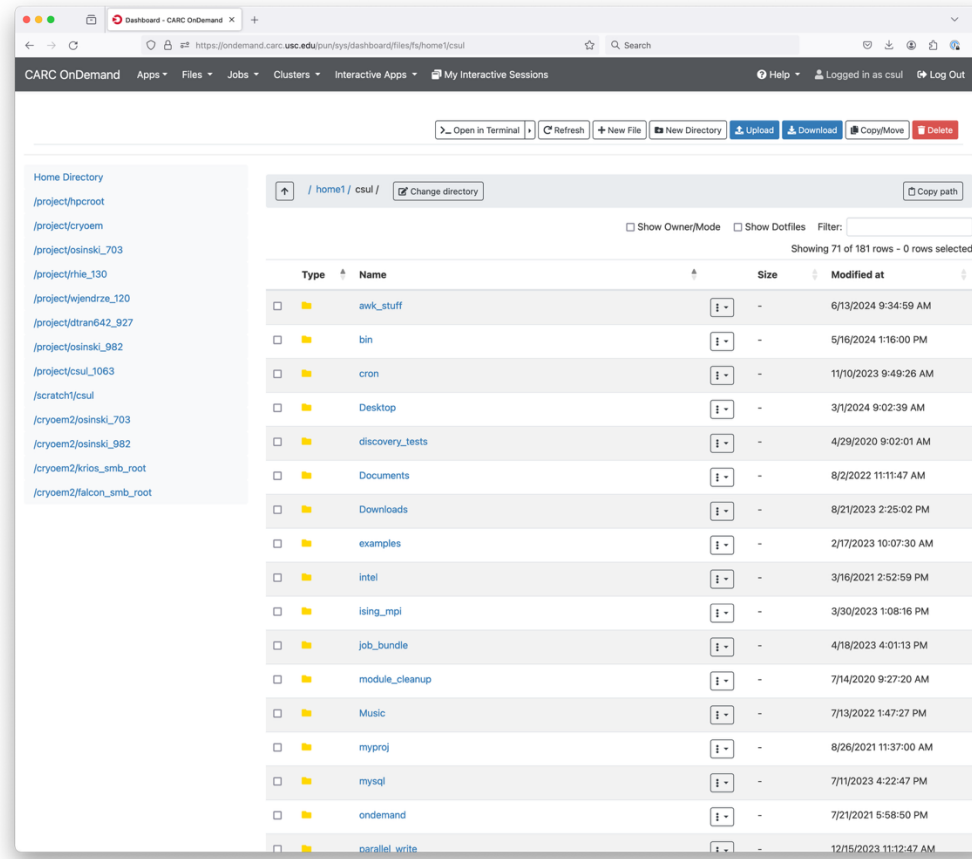
USC

| Advanced Research Computing
Enabling scientific breakthroughs at scale

Uploading Data

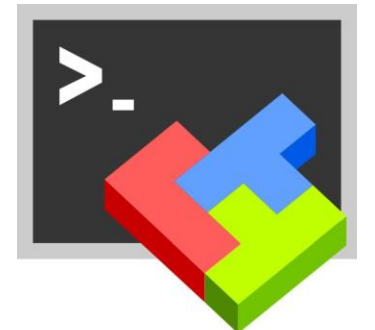
- There are many ways to get data onto CARC
- We will discuss
 - Open OnDemand
 - Command line (rsync, scp)
 - SFTP Clients (CyberDuck, Filezilla)
 - Globus (Files at other HPC Centers)

Open OnDemand



SFTP Client

- Secure File Transfer Protocol
- Access files with GUI Interface
- Separate App
- Documentation page [here](#)
- Hostname: hpc-transfer1/2 or discovery.usc.edu



rsync

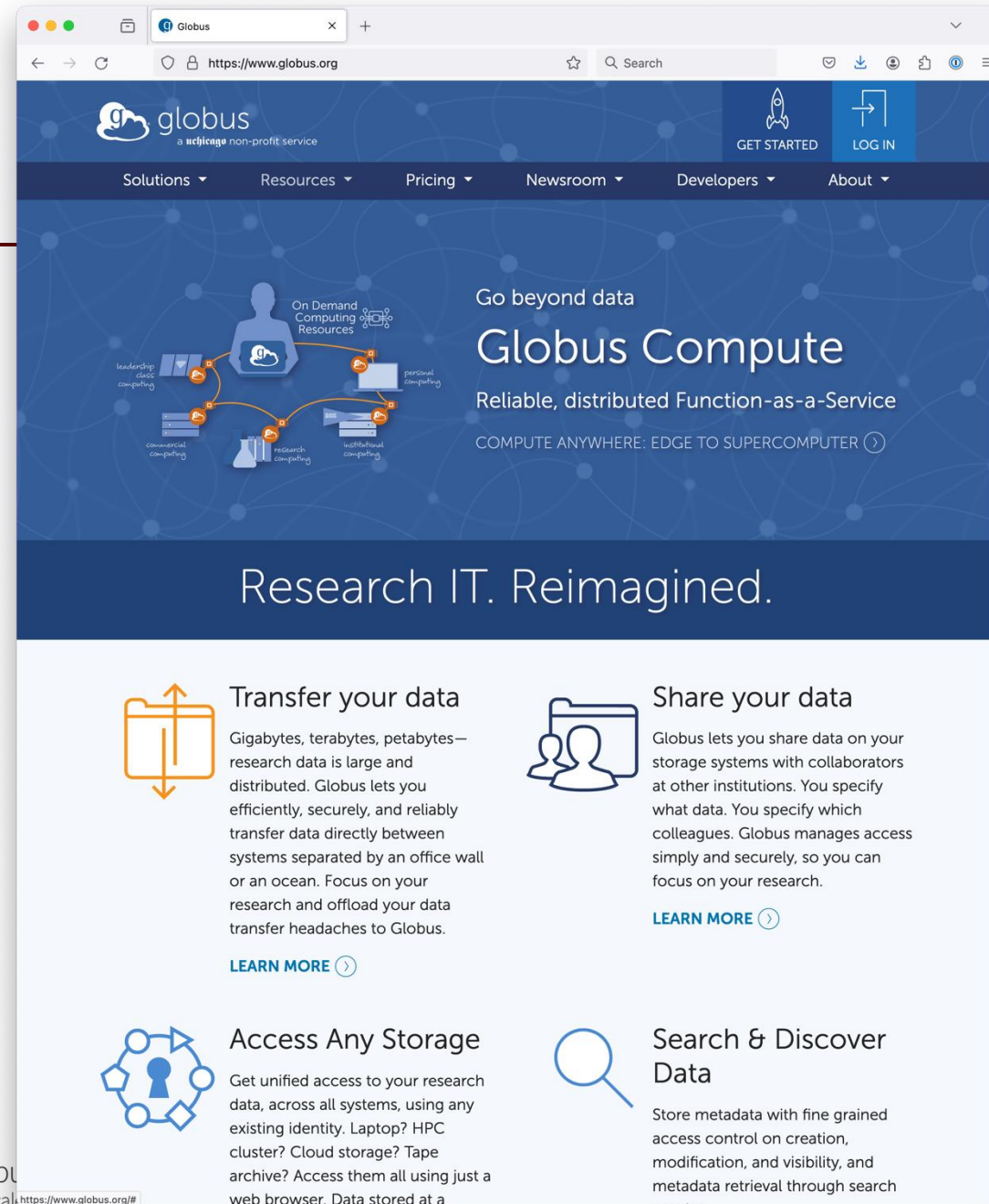
- To copy between computers rsync is easiest
- Authentication credentials required
- Transfers can be resumed if interrupted
- Works across filesystems too

```
rsync -rltvh <source> <target>
```

```
$rsync -rltvh ~/Data/ ttroj@discovery.usc.edu:/project1/ttroj/Data
```

- ~/Data/ will copy directory contents while ~/Data will copy directory

Globus



The screenshot shows the Globus website homepage. The browser address bar displays "https://www.globus.org". The header includes the Globus logo, navigation links (Solutions, Resources, Pricing, Newsroom, Developers, About), and buttons for "GET STARTED" and "LOG IN". The main banner features a diagram of "On Demand Computing Resources" with icons for leadership, commercial, research, and institutional computing, alongside the text "Go beyond data", "Globus Compute", "Reliable, distributed Function-as-a-Service", and "COMPUTE ANYWHERE: EDGE TO SUPERCOMPUTER". Below the banner, four sections are highlighted: "Transfer your data" (Gigabytes, terabytes, petabytes—research data is large and distributed. Globus lets you efficiently, securely, and reliably transfer data directly between systems separated by an office wall or an ocean. Focus on your research and offload your data transfer headaches to Globus. LEARN MORE), "Share your data" (Globus lets you share data on your storage systems with collaborators at other institutions. You specify what data. You specify which colleagues. Globus manages access simply and securely, so you can focus on your research. LEARN MORE), "Access Any Storage" (Get unified access to your research data, across all systems, using any existing identity. Laptop? HPC cluster? Cloud storage? Tape archive? Access them all using just a web browser. Data stored at a), and "Search & Discover Data" (Store metadata with fine grained access control on creation, modification, and visibility, and metadata retrieval through search).



Advanced Research Computing
Enabling scientific breakthroughs at scale

<https://www.globus.org/#>

File Manager | Globus

https://app.globus.org/file-manager?origin_id=56569ec1-af41-...

Search

File Manager

Panels

Collection USC CARC User Directories

Path /home1/csul/

select all up one folder refresh list filter view

NAME	LAST MODIFIED	SIZE
andrewds_cpu_req	6/14/2024, ...	17.91 K
andrewds_mem	6/14/2024, ...	91 B
andrewds_mem_req	6/14/2024, ...	1.50 K
andrewds_p_and_q	6/14/2024, ...	15.61 K
andrewds_sprio	6/14/2024, ...	34.14 h
awk_stuff	6/13/2024, ...	-
bin	5/16/2024, ...	-
bisection_test.ipynb	12/6/2022, ...	45.51 h
cf_data_test	4/4/2024, 1...	6.97 K
configurator.html	3/19/2024, ...	31.49 h
cron	11/10/2023,...	-
cronlog.txt	2/14/2024, ...	0 B
dashboard.json	4/10/2024, ...	24.17 K
data_managment.zip	6/19/2023, ...	10.10 h
Desktop	3/1/2024, 0...	-
discovery_tests	4/29/2020, ...	-
Documents	8/2/2022, 1...	-
Downloads	8/21/2023, ...	-
examples	2/17/2023, 1...	-

Share

Transfer or Sync to...

New Folder

Rename

Delete Selected

Download

Open

Upload

Get Link

Show Hidden Items

Manage Consent

Globus

- Similar user interface to Open OnDemand
- Fast connection to other HPC Centers
- Share data with external collaborators

Getting Help

- Request assistance
 - Email carc-support@usc.edu
 - Office Hours (drop-in)
 - Every Tuesday@2:30pm Zoom & LVL 3L
- Learn more!
 - Visit carc.usc.edu
 - Request a consultation (anytime)
 - Attend a Workshop (when scheduled)
 - [Visit our Discourse page!](#)



Thank you for attending!

Questions?

carc-support@USC.EDU



USC

Advanced Research Computing
Enabling scientific breakthroughs at scale